CSC 244 — FinGuard: Installation and Demonstration Guide

Team number: **TeamXX (replace)** | Project: A Role-Based Multi-Agent System for Collaborative Financial Fraud Detection | November 2026

This guide describes the runnable implementation distributed with the source ZIP. It is intentionally a setup/readme rather than a research report. The project uses only synthetic demonstration records.

1. Architecture and present capabilities

```
Synthetic transactions → SQLite evidence queries
          ↓
Employee | Manager | CFO | Fraud Analysis
          ↓
Coordinator consensus (LOW / MEDIUM / HIGH)
          ↓
Local browser dashboard → human accountant disposition
          ↓
Saved review runs + agent messages + audit history
```

The system uses deterministic role-specific Python functions, an SQLite relational store, a custom standard-library HTTP server and a browser dashboard. It does not use LLMs. The PostgreSQL schema is proposed only; LangGraph, true second-round debates, real accounting imports and hosted deployment are not implemented.

Review outcome is a priority recommendation, not a finding that fraud occurred. All case labels and policy thresholds are illustrative.

2. Prerequisites and Windows installation

Required: Windows 10/11 or comparable operating system, Python 3.9+, modern Chrome or Edge. Git is required only for repository collaboration. No pip packages, API keys, paid services or external database are needed.

Option A — download and extract **TeamXX_code.zip**, then open PowerShell in the extracted root containing `app.py`.

```
py -3 scripts/run_tests.py
py -3 app.py
```

Open **http://127.0.0.1:8501**. Leave the PowerShell window running and stop with Ctrl+C. If `py` is unavailable, substitute `python` or `python3`.

Option B — clone an existing GitHub repository, replacing placeholders:

```
git clone https://github.com/YOUR-ACCOUNT/YOUR-REPO.git
cd YOUR-REPO
py -3 scripts/run_tests.py
py -3 app.py
```

Automatic initialization creates `dashboard_demo.sqlite` locally with six synthetic records and initial persisted risk assessments. This database is deliberately excluded from version control.

Windows shortcut and troubleshooting

You may launch with `.\launch_dashboard.bat`. If the batch script fails or the terminal closes, prefer the visible command `py -3 app.py`. If port 8501 is occupied, use `py -3 app.py --port 8502` and browse to port 8502.

3. Five-minute functional demonstration

Step	Action and expected result
1. Dashboard	Open the browser. Verify six seeded transactions: 3 HIGH, 2 MEDIUM, 1 LOW.
2. Suspicious payment	Select F-001. Review lack of controls, invoice mismatch and agent assessments.
3. Split transactions	Select S-001. Compare S-001 ($4,800) + S-002 ($4,700); split_group G-001 totals $9,500.
4. Contrast	Select L-001 to show a normal payment and LOW-risk recommendation.
5. New review	Click Run agent review and verify a new review run appears in history.
6. Human review	Save Request additional evidence with a note. Refresh to confirm persistence.

4. Executable SQL evidence

Open a second PowerShell session in the same directory. Run:

```
py -3 show_audit.py
```

The script executes a transaction/employee/vendor JOIN; suspicious split GROUP BY/HAVING; most recent consensus; per-agent F-001 evidence; and saved accountant decisions. These are real queries against the SQLite audit database, not pre-rendered outputs.

If the SQL output lists no human review, complete step 6 above first. You may also run `py -3 prototype.py` for the standalone command-line output.

5. Automated tests and expected evidence

Suite	Count	Checks
test_prototype.py	5	Risk cases and presence of four assessments, four messages, one consensus
test_app.py	5	Dashboard counters, split aggregate, reruns, human persistence, invalid input
test_http_api.py	5	HTTP GET, HTML, review POST, human POST, 400 error handling
TOTAL	15	All 15 passed in the compiled package verification, November 9, 2026.

```
py -3 scripts/run_tests.py
```

A successful test run prints **Ran 15 tests ... OK**, returns exit code 0, and saves the exact log under `.reports/tests.txt`. The original uploaded ZIP had 10 passing tests; five HTTP tests were added during compilation. Every test uses in-memory/temporary SQLite data and the HTTP tests use an ephemeral localhost port.

GitHub Actions runs the suite for Python 3.10 and 3.12 on Ubuntu and Windows after pushes and pull requests. Actions is a test runner and does not host a public dashboard.

Tests establish deterministic behavior on six seeded records; they do not demonstrate broad fraud-detection accuracy, financial compliance, robust security, or production deployment.

6. GitHub repository and team evidence

Create an **empty private** repository at `https://github.com/new` and avoid auto-generating README or .gitignore. In the extracted source root:

```
git init
git branch -M main
git add .
git commit -m "Import tested reference baseline"
git remote add origin https://github.com/YOUR-ACCOUNT/YOUR-REPO.git
git push -u origin main
```

Repo Settings → Collaborators → Add people, invite your partner. Each person should create a feature branch, commit their own actual work, open a PR and demonstrate passing Actions checks. Git auth may require a browser login/credential manager or personal access token; a normal GitHub account password will not work for HTTPS pushes.

```
git switch -c feature/partner-implementation
# work on actual files and run tests
py -3 scripts/run_tests.py
git add .
git commit -m "Implement and test partner changes"
git push -u origin feature/partner-implementation
```

For individual evidence, use `git log --graph --oneline --all`, `git show --stat <commit>`, demo screenshots and saved test logs. Finish `docs/TEAM_CONTRIBUTIONS_TEMPLATE.md` with truthful task ownership and percentages.

The ZIP supplied for compilation did not contain verifiable per-student authorship or separate partner test files. Do not assign its tests to either person without corroborating commits or confirmation. Under the course project guide, each member of a two-person team must contribute at least 45% of the phase.

Canvas submission checklist

TeamXX_presentation.pptx — slides with verified evidence; **TeamXX_code.zip** — commented source and GitHub workflow; **TeamXX_readme.pdf** — this installation guide. Rename XX to the actual team number, and reconcile names/contributions before submission.